GSE 518
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Problem Set 9

You must turn in problems: 1, 2, and 3(a). However, you are strongly encouraged to work on all the problems before the final exam, since the material they cover will be part of it. Please let me know if you have any questions.

1. Let $X_1, X_2, \ldots, X_n$ be a random sample from a normal distribution with *known* mean 0 and *unknown* variance σ_0^2, $N(0, \sigma_0^2)$. Two possible estimators of σ_0^2 are:

$$v^2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2 \quad \text{and} \quad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2$$

   where as usual $\overline{X}$ is the sample average.

   (a) Are these estimators unbiased?

   (b) Which estimator has lower mean square error? (*Hint:* You can take as given that $\frac{nv^2}{\sigma_0^2} \sim \chi^2_{(n)}$, and that a chi-squared distributed random variable with r degrees of freedom has mean r and variance $2r$. You may want to check question 3 from Problem Set 7.)

2. Let $X_1, X_2, \ldots, X_n$ be a random sample from each of the distributions having the following probability density/mass functions. In each case answer the following questions: (i) find the Cramèr-Rao Lower Bound for any unbiased estimator $\widehat{\theta}$ of θ_0; (ii) Is $\widehat{\theta}_{MLE}$ a Uniform Minimum Variance Unbiased Estimator (UMVUE) in each case?

   (a) $f(x|\theta_0) = \frac{\theta_0^x e^{-\theta_0}}{x!}$, for $x = 0, 1, \ldots$ and zero otherwise, for $\theta_0 \geq 0$. (*Note:* this is the pmf of a Poisson distribution, so $E(X_i) = \theta_0$ and $Var(X_i) = \theta_0$).

   (b) $f(x|\theta_0) = \frac{1}{\theta_0} e^{-(x/\theta_0)}$, for $0 < x < \infty$, and zero otherwise, for $0 < \theta_0 < \infty$. (*Note:* this is the pdf of an exponential distribution, so $E(X_i) = \theta_0$ and $Var(X_i) = \theta_0^2$).

3. Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with probability density function

$$f(x|\theta_0) = \frac{2x}{\theta_0^2}, \text{ for } 0 < x < \theta_0, \text{ and zero otherwise}$$

   (a) Show that $\widehat{\theta}_1 = \frac{3\overline{X}}{2}$ is an unbiased estimator for θ_0.

   (b) Find the Cramèr-Rao lower bound for any unbiased estimator $\widehat{\theta}$ of θ_0. For this particular problem, use the following formula to calculate the CRLB: $CRLB = 1/E\left[S^2(X, \theta_0)\right]$. (*Note:* In general, you could use any of the three formulas we saw in class to calculate the CRLB –and, usually, which one is easier to use depends on the particular problem. However, as explained below, in this particular problem the assumptions of the CRLB and the Information Matrix Identity do not hold, so each formula can give a different result.)

(c) Show that the variance of $\widehat{\theta}_1$ is *less* than the Cramèr-Rao lower bound.

Note for general knowledge: One of the assumptions for the Cramèr-Rao lower bound to hold, and for the score and information matrix identities, is that we can interchange the order of differentiation and integration (e.g., see derivation of these results in CB). When the range of the random variable depends on the parameter, this assumption is not satisfied and those results no longer hold. In this problem this is the case, since the pdf is positive for $0 < x < \theta_0$.)